Innovation in mining – a contractor's role

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ABSTRACT

Underground contractors in the mining industry must be prepared to take a position at the forefront of innovation of equipment and work methods in order to maintain a competitive advantage over in-house operations and remain relevant. Innovative concepts such as split-feed development jumbos, high capacity trucks and loader guidance systems were implemented by contractors to stay ahead of 'the pack'. All of these are now common across the industry.

The demands of regular participation in competitive tendering coupled with aggressive fleet depreciation, means contractors tend to have more communication and opportunities to work with equipment suppliers than mine owners. Therefore, they are more likely to influence equipment design and product directions. They also have a unique opportunity to assist in trialling new concepts and technologies across a wide variety of operating conditions rather than being restricted to one mine.

In addition, as a 'guest' on a mine site, mining contractors do not have the luxury of creating the environment for their preferred systems, rather they must have the capacity to adapt their technology to suit the offered environment at a particular mine. This leads to a pursuit of adaptive technologies that can be implemented selectively and seamlessly without excessive disturbance rather than 'all-of-mine' systems. These incremental innovations then become available for use across the mining industry where they can add value as 'bolt-on systems' to existing operations.

Mining contractors, therefore, act as the bellwether for industry practices and equipment. However, in order to maintain their position they must engage industry stakeholders to deliver practical, innovative and cost-effective technologies that add value and result in an improved bottom line for all. This paper uses case study material to illustrate how the Byrnecut Group has approached innovation and how the theory of Disruptive Technology applies to underground mining innovation.

INTRODUCTION

Underground mine contracting is a challenging environment to work in. Not only is there competition within the sector, but there is competition with mine owners considering an owner-miner solution. In order to remain competitive, contractors have had to evolve and 'value-add' to an operation through more than just attractive rates and increased efficiencies. The role of a contractor has changed significantly since the 1980s West Australian gold boom. Where once a contractor was evaluated purely on cost and operational capability, current contract evaluation has expanded to assess occupational health and safety, training, quality, environmental standards, Innovative cultures and application of technology.

Professor Clayton M Christensen of the Harvard Business School introduced the concept of Disruptive Innovation and Disruptive Technology, which investigated the failure of large, successful businesses and what had caused their failure (Lambert, 2014). His theory identifies 'sustaining technologies' and 'disruptive ones' which categorise the way a product or concept is delivered and maintained in a market. Sustaining technologies refer to continuous improvement through process, quality and/or technology of an existing product that is well established and accepted in a market. Disruptive technology was defined in *The Innovator's Dilemma* as 'innovations that result in *worse* product performance, at least in the near term'(Christensen, 1997). This theory has application to the mine contracting sector, as contractors need to adapt to a pre-existing environment and existing 'off-the-shelf' solutions may not be practicable.

This has resulted in contractors taking the lead in innovative and disruptive technology to gain a competitive edge against both their peers and mining houses. Due to the nature of the business, and frequent competitive tenders (Byrnecut Australia tendered 17 projects in 2016), each advantage is quickly eroded and superseded by the 'next' development as each innovation becomes adopted as the new standard. In addition, each project has its own unique requirements, resulting in an environment where 'one size' definitely does not 'fit all'. This paper will focus on the role innovation plays within a contractors business.

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TRENDS IN UNDERGROUND MINING

Underground metalliferous mining in Australia, and specifically the requirements of recently tendered projects have been along four main themes:

1. deeper
2. bigger
3. smarter
4. cheaper.

Whilst greenfields projects are relatively uncommon, existing mines are expanding by developing deeper orebodies and increasing production rates through larger and higher capacity trucks and loaders. Targeted advance rates are increasing, with varying degrees of success, which is driving improvements in efficiencies and applications of technology. Whilst capital costs and labour rates are increasing, the only option to deliver cost savings is through embracing innovation!

Deeper

As mines have increased in depth, travel times to and from the production areas have increased leading to less productive time per working shift. An example is the Gwalia mine in Western Australia where mine development is currently 1700 m below surface and accessed only by decline. In order to maintain acceptable rates of production, the mine has had to overcome the logistical issues of transporting personnel long distances and loss of productive time.

Scheduling and planning can reduce operational downtime, however, technology, in the form of automation has improved productivity by allowing machinery to continue to operate during an operator's travel time or operational breaks.

An example of the role of a contractor is implementing this technology is provided below.

Bigger and smarter

During the course of the author's career, there has been an increase in capacity and capability in drilling, loading and hauling equipment:

- development jumbos have increased from 5.37m (12') feed lengths to 7.2 m (18') feed lengths resulting in advance of up to 5 m per cut and have the capability for autonomous drilling capability
- loaders have increased in capacity from 14–16 t (Elphinstone R2800 and Toro 501) up to 20–21 t (Caterpillar R3000 and Sandvik LH621) and are now commonly operating autonomously
- trucks have increased from 40 t capacity (Elphinstone AD40 and Toro 40D) up to 65 t (Atlas Copco MT65).

In order to maintain existing productivity levels as mines become deeper, or improve on historic productivity levels, the mining industry has embraced automation. An example is that of production loaders, where once 'line of sight' remote operation was common practice, it has all but vanished to be replaced with teleremote control and various forms of automation.

Loader automation has also resulted in improved productivity above that anticipated during implementation. When the Caterpillar MineGem system was introduced to the Newmont Jundee operations in 2007, Caterpillar R1700G loaders were being used to teleremote bog the stopes. At the time, teleremote loaders were speed restricted to tramming in 1st gear as higher speeds resulted in machine damage. In this application, a productivity rate of 50 t/hr was being achieved. With the implementation of automation, the loader was able to be operated at higher speed in the same drive, achieving higher productivity rates. However, due to the sophisticated navigation and tracking system, the mine was able to use a larger Caterpillar R2900G loader in the same envelope further increasing productivity to 75 t/hr.

Production loader automation has reached a sustainable level and is continuously improving to the point where recent installations are trialling auto-dig functionality along with existing tramming options.

Loader automation has now been implemented into development headings to improve advance rates and minimise heading downtime due to blasting re-entry (International Mining, 2017). For development applications, loader automation is still a disruptive technology in that it has been trialled, but is not completely reliable or applicable to all operations, rather being specific to niche high priority development.

The main hurdles to successful implementation and uptake of development automation are outlined below:

- Current automation systems rely on a mine map for the loader to navigate the work area, and with development resulting in an expanding work area rather than a retreating work area, this map requires continuous updating for the loader to function. Newer systems that allow a machine to self-navigate and learn the tunnel will address this constraint.
- Dust generated by blasting affects laser barriers, scanning lasers and cameras. Laser barriers can be replaced by physical barriers (gates, lanyards etc), however a short period of time is still required to allow airborne dust to settle before attempting to operate the loader. Automatic dust suppression sprays can reduce the time but also require installation, maintenance and rely on personnel to 'arm' the system. Depending on the re-entry period for the heading, there is limited potential gain. For a single heading decline development with long stretches of secondary ventilation and lengthy re-entry periods, autonomous loaders become attractive.
- The mine schedule dictates the priority work areas and few mines have a single high priority development heading. Therefore, the potential to increase development in one location in the mine is often not required, and maintaining multiple headings with the required hardware for autonomous loading is often uneconomic.
- The current system of manually setting up barriers, positioning loaders within the automation zones and maintaining navigation systems is heavily reliant on 'buy in' from operators and support from supervisors and mine planners to maximise effectiveness of the system. In the hectic final moments of preparation for a blast, setting up for autonomous loading is not a default process for the charge crew.

As automation systems become increasingly sophisticated and are able to learn and generate their own mine maps, along with the introduction of complete mine communication networks which do not require expensive automation hardware, development loader automation will become the norm, as it has become for production.

ROLE OF A CONTRACTOR

So, where does a contractor fit in?

During the 1990s, major mining houses such as Western Mining, BHP and Mt Isa Mines maintained engineering departments to develop solutions to site specific problems

and conducted the majority of engineering projects 'in-house'. As company philosophy has changed and trends such as centralising, globalising and out-sourcing have been implemented, this has led to the demise of 'in-house' engineering.

As stated by Bryant, 'many mining companies do not have the knowledge or resources to implement dramatic technological solutions' (Bryant, 2015) and whilst it is outside the scope of this paper to discuss the reasons behind this change, it provides an opportunity to the contractor to provide a resource to the mining company.

Contractors need to adapt to different work environments, geological conditions, infrastructure, mine sizes and geographical locations. They do not own the mine, have control over the mine design, infrastructure or longevity, and are governed by the site specific terms and conditions which will vary across contractor projects. This necessitates a flexible mind set to be able to offer 'all things to all people' and requires continuous improvement and monitoring of industry benchmarks to remain viable.

Contractors do not have the luxury to design a mine around an automation system or specific equipment philosophy and therefore tend to adapt incremental or 'bolt-on' systems that can be upgraded or upscaled to suit the needs of multiple clients or adapted to work within given mine constraints.

This links to the theory of Disruptive Technology, where a contractor can take a risk on the performance of alternative technologies, with the intent to develop a sustainable, reliable product through the assistance of both the original equipment manufacturer and the mine owner.

Original equipment manufacturers engagement

Due to the regular engagement with original equipment manufacturers (OEMs), contractors have a better opportunity to influence equipment design and application than irregular customers, for example, Byrnecut are regularly one of Sandvik Mining's annual top 5 global customers (Mauger, personal correspondence, 2017). A direct result of this engagement is the increase in capacity of mining equipment. As outlined by Wood, contractors have the ability to engage with an OEM to develop equipment for the industry which can address the requirements of multiple mine sites with the experience obtained through operating on various projects rather than developing project specific machines with limited application to the wider mining community (Wood, 2014).

To develop a competitive tender, contractors assess the physical targets required and determine the most suitable fleet and manning levels to achieve these targets. They will often tender these requirements to various OEMs for options and prices. The tender process therefore flows on to the OEMs and encourages competition, improvement and development within the sector. The examples below illustrate this rate of development, industry acceptance and transition from disruptive to sustainable.

Truck development

As previously stated, there has been a large increase in truck haulage capacity over the previous 20 years. Table 1 shows the increase in 'standard' underground haulage trucks over this period. Each of these trucks is used in the same 5.5 m wide decline, but capacities, power and productivity have rapidly increased to improve mine production. Individual truck productivity has increased from 55 000 tkm per month per truck to 100 000 tkm per month whilst still operating within the same mine decline. Regular competitive tendering of existing mines has resulted in trucking fleets regularly being replaced/upgraded as each contractor attempts to gain an edge on their competition enabling the mine owner to take advantage of product development more rapidly than otherwise may have been possible. Anecdotally, the Caterpillar AD55 was only developed at the request of a well known mining contractor (Watkinson, personal correspondence, 2015).

TABLE 1
Articulated truck development.

Manufacturer	Model	Year	Capacity (t)
Sandvik	Toro 40D	1991	40
Caterpillar	AD40	1994	40
Sandvik	Toro 50D	1995	50
Atlas Copco	MT5000	1996	50
Caterpillar	AD55	2001	55
Atlas Copco	MT5010	2001	50
Sandvik	Toro 50+	2004	50
Sandvik	Toro 60	2005	60
Caterpillar	AD55B	2007	55
Atlas Copco	MT6020	2014	60
Caterpillar	AD60	2014	60
Sandvik	TH663	2015	63
Atlas Copco	MT65	2017	65

Boxhole development

In raise bore shafts, the longest component of the cycle is the drilling and reaming activity, and a raise bore machine is selected for a project based on the theoretical penetration rate in the ground conditions (Wood, 2014). However, for shorter boxhole rises, the overall construction cycle is weighted towards the non-drilling activities. Therefore, to optimise the boxhole construction process, the bottleneck in the process is not the drilling speed, but the set-up, rig up, tramming and other non-drilling activities (Salo, 2016).

In 2011, Raising Australia commenced development of a mobile boxhole drill (Figure 1), which could drill holes to 750 mm diameter and at an angle of up to 45° from vertical. This machine evolved into the Weller Boxhole drill and has been delivered to the Jundee mine site for slot drilling works. Whilst the prototype machine has ultimately achieved its goal of being a mobile, mechanical slot drilling alternative to hand held ladder rising, it had room for improvement and further development.

The original prototype was designed to coverall angles from 45° to 90° which resulted in additional support, bracing and more cumbersome drilling frame than necessary. Following



FIG 1 – Weller Boxhole.

the trials of the prototype Weller Boxhole drill, a sample of 84 slots drilled at Jundee was analysed to develop a realistic operating range for a new boxhole machine. This sample resulted in an average hole length of 15.44 m and average hole angle of 73.73°. Assessing the quantity of holes drilled across brackets of 5° angles identified that 93 per cent of all holes required an angle of up to 60° from horizontal (30° from vertical) – refer Figure 2.

A time and motion study carried out on the Weller Boxhole rig identified the following areas as being bottlenecks in the overall slot drilling cycle:

- tramming
- installing muck and transfer chutes
- removing cuttings
- installing stabilisers
- pulling rods.

As the intention for a mobile slot drilling machine is to deliver a bored slot with the least reliance on and disruption to the mining operations as practicable, it was deemed imperative that the next machine should be as self sufficient as possible, with a target of being able to deliver a 20 m slot within a 24 hour period, inclusive of machine set-up and de-rig.

Through a tender process carried out in 2015 to procure a new boxhole machine, several manufacturers were requested to provide an offer for a mobile slot drilling machine.

TRB-Raise borers (TRB) had recently developed the Rhino 100HM concept as a highly mobile, high speed solution to the entire slot construction cycle in response to a specific set of requirements from the Agnico-Eagle Kittila mine. Whilst this machine had been custom designed to suit the requirements of a single operation, further development was required to make it a mainstream alternative to existing slot opening solutions.

The Rhino 100HM embodies all the attributes of disruptive innovation:

- solves all the requirements of a specific problem
- assesses the entire problem, not one component of the problem
- addresses an existing bottleneck
- offers reduced performance (660 mm slot) compared to current options (750–1500 mm).

The Rhino was identified as offering the best concept whilst not addressing all the required specifications, namely the ability to drill angled holes and drill holes up to 750 mm in diameter. However, as this machine addressed the majority of requirements TRB was engaged to further develop the existing Rhino concept to be suitable for the Australian requirements. Using the historical records of hole lengths and geometry from the Jundee mine, along with questionnaires sent to existing clients to define the most suitable range of hole size and geometry, Raising Australia was able to provide sufficient data to TRB to redesign the drilling module to suit these requirements (refer Figure 3).

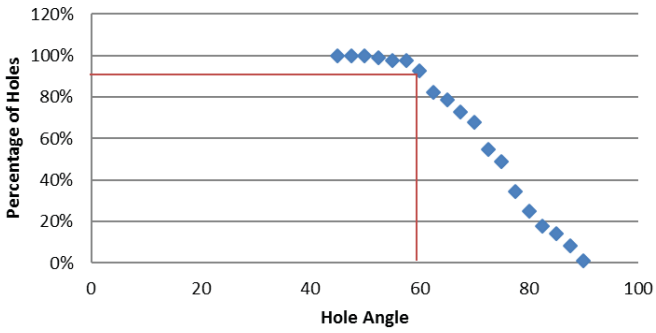


FIG 2 – Boxhole distribution by angle.

Results from the Kittila mine indicate that the performance and capability of the Rhino 100HM boxhole machine have led to a change in mine scheduling as slot drilling is no longer a critical path activity. Slots can be constructed on-demand and in the event of a stope failure, re-slotting results in a two day delay to production, allowing greater flexibility in mine planning and scheduling.

The first Rhino 100HM will be delivered to an Australian operation in September 2017. As a disruptive innovation, this machine does not offer boxhole sizes as large as current machines, however the overall cycle improvements and the operational flexibility offered by this system will challenge traditional stope design and scheduling to allow increased productivity and equipment utilisation. It has taken significant contractor engagement to bring this bespoke mining solution to the wider industry.

Innovation, research and development

In addition to striving to provide the most efficient and therefore profitable service to mine owners, contractors also have a unique opportunity to conduct in-house research and development with diversified test sites and proving grounds at hand. We cannot rely on manufacturers, suppliers and mining companies to know what's best for our business and have a responsibility to lead the way.

Whilst all great inventions commence with a good idea, not all great ideas solve a problem! As teachers and lecturers regularly bemoan of students, 'Answer the question'.

Therefore, the approach taken by the Byrnecut Group in approaching innovation, research and development activities is to first 'define the question', which in turn leads to an effective and practical solution.

To illustrate, some examples of innovation and product development activities undertaken by the Byrnecut Group are outlined below in line with this philosophy.

Paste fill reticulation diverter

Background

Paste fill reticulation for underground mines is increasing in complexity as mines become deeper and the need to rapidly transition from one stope to the next is a critical path on the mine production schedule. This, coupled with a mandate to minimise the operating costs through binder expenditure, drives operations to walk a fine line on solids concentrations which makes blockage/stoppage somewhat inevitable.



FIG 3 – Rhino 100HM boxhole.

Historically, activities such as clearing blockages, flushing lines and changing fill destinations have been manually intensive and risky tasks. Reticulation, especially valving, used in paste fill systems has been adapted from other industries, particularly wastewater, and does not address the specific requirements of paste transfer and is often over engineered for its end use.

Question

Can a robust, simple valve arrangement be designed which can handle the pressure, abrasivity and flow rates of paste fill in DN200 or DN250 steel pipe?

Solution

Quattro Project Engineering has developed a series of automated diverters specifically for paste filling duties.

The 2-position QD2™ is a small, compact self contained diverter capable of mounting to the backs and actuates via standard mine air that can divert the paste reticulation between two outlets, such as:

- stope 1 and stope 2
- stope and dump (flush)
- stope and main reticulation line.

The 4-position QD4™ is a larger frame mounted diverter offering up to four possible positions to enable the following actions:

1. outlet to stope 1
2. outlet to stope 2
3. outlet to dump (flush)
4. isolation (manual and electric).

The isolation position blocks off the inlet line enabling positive isolation for maintenance activities and ensures maximum pipe pressure cannot be exceeded in the case of a downstream blockage.

Both diverters can be controlled remotely from the paste plant or other control room and enable the operator to efficiently manage the flow of paste within the system and provide safe and fast reaction to blockage when integrated with strategic instrumentation (pressure sensors), thus protecting the reticulation from damage and reducing the risk of production delays (Figure 4).

Elevated work platform

Background

The release of Mines Safety Bulletin No 114 regarding 'Compliance requirements for multipurpose plant' (Department of Mines and Petroleum, 2014) required work platforms used with Integrated Tool Carriers to conform to the requirements of AS 1418.10:2011 Cranes, hoists and winches – Mobile elevating work platforms.



FIG 4 – QD4™ paste valve.

Traditional work baskets were not designed in accordance with this standard and would not be permitted for use as an elevated work platform. Elevated work platforms designed in accordance with the relevant codes were intended for the surface construction environment and did not have the durability required for underground mining. There was a requirement for a work basket to suit an Integrated Tool Carrier that would conform with the requirements of AS1418.10 without affecting the practicality and flexibility of a multi purpose IT.

Question

Can a simple modification be made to an existing work basket and IT to ensure it conforms with the required codes?

Solution

Byrnecut engaged Quattro Project Engineering (QPE) to develop a work basket that would conform to the code requirements.

Extensive consultation with the DMP and Worksafe in conjunction with internal risk assessment, gap analysis to AS1418.10, the formulation and implementation of a suitable test plan resulted in a conforming work platform design which has since been manufactured.

A design registration was approved for the required work platforms by Worksafe enabling continued use of IT work platforms whilst conforming to the latest safety requirements.

Fan handling platform

Background

Hanging secondary ventilation fans is a time consuming, manual task exposing employees to hazardous and cramped work conditions.

Question

Can a system be designed that utilises existing equipment where possible (ie Integrated Tool Carrier) to allow mechanical fan installation without the need for expensive single application equipment?

Solution

To minimise exposure of personnel to the hazards of working at height and working in a constrained environment, a system to remotely install the fans using a purpose built handling platform and guided locking bracket to support the fan has been designed. The fan can be installed and removed by the IT operator, without requiring personnel in the basket. A camera and lighting system with a display in the IT cab provide vision to the fan bracket during installation, and the IT operator has control over the basket gripper arms to release the fan when the installation is complete.

The entire structure including hanging bracket rock bolts has been engineered in accordance with relevant Australian Standards and state mining legislation.

Figure 5 shows the fan assembled on the stage, and Figure 6 shows how to install the fan onto the hanging bracket. Due to the overall length and weight of the fan, the outlet silencer is removed for fan installation, then attached to the fan using the same process.

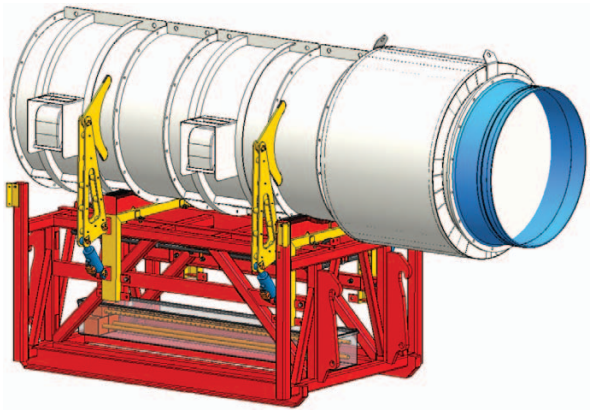


FIG 5 – Fan handling platform.

Ejector truck

Background

Ejector trucks are commonly used in mine backfilling applications, however, traditional designs have resulted in reduced payload compared to tip trays and are often maintenance intensive.

Question

Can an ejector truck system be designed with payload comparable to a tip tray?

Solution

In conjunction with Transmin Pty Ltd, an alternative underground ejector truck based on the Low Profile Feeder technology is currently being developed. The LPF system is a hybrid design combining elements from belt and apron feeders to enable a heavy duty belt capable of sustaining impact loads from blasted rock loaded into the truck whilst occupying a minimal envelope at the base of the tray.

The proposed ejector tray has been developed to fit on an Atlas Copco MT6020 haul truck and provide a useable payload of 30.3 m³ (SAE Heaped).

This system has been proven in fixed plant applications and offers a higher capacity payload than existing ejector concepts. The prototype is under way and will be trialled mid 2017 at a Western Australian mine (Figure 7).

Shaft sinking equipment

The Byrnecut group has extensive experience of shaft sinking in the Asia-Pacific region, and most recently completed the Ernest Henry Haulage shaft for Glencore plc at the Ernest Henry Copper mine near Cloncurry, Queensland Australia. Applying our lateral development philosophy of innovation and technology to vertical development, Byrnecut have developed and enhanced the methodology for constructing

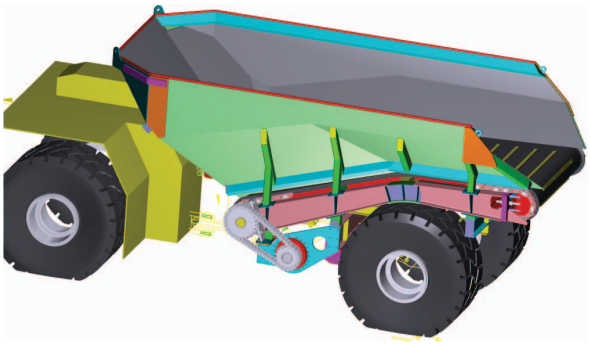


FIG 7 – Low profile feeder ejector truck.

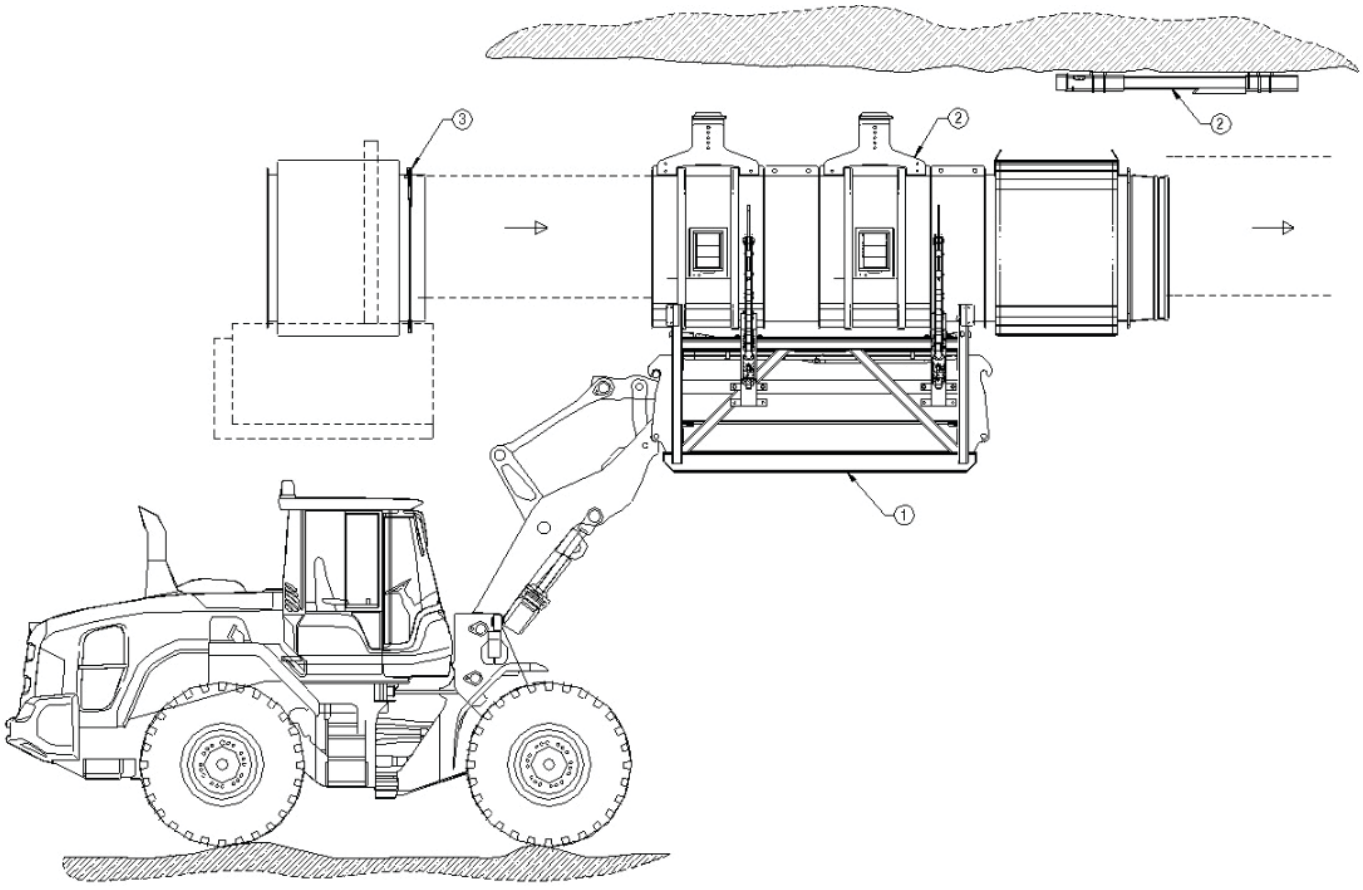


FIG 6 – Fan installation sequence.

vertical shafts in the region. Our guiding philosophy has been to minimise exposure to personnel by embracing developments in technology and engaging with clients and consultants to facilitate a safe and efficient shaft construction. A number of innovations have been introduced to shaft sinking and are outlined below:

Radio remote controlled shaft jumbo with proximity detection

Background

A traditional shaft jumbo is operated from control boxes mounted on the feed rails, requiring the operator to be close to the revolving drill string, or by remote controls and umbilical cables that introduce additional hazards. Personnel are exposed to the hazard of the energised drill string and concerns have been raised regarding personnel being ion the bench whilst the jumbo is drilling.

Question

Can the jumbo be operated remotely and incorporate an interlock to shutdown the drill if personnel are within a hazardous zone?

Solution

In conjunction with Bucyrus (then Caterpillar), a radio remote control system was developed and commissioned for operation of the shaft jumbo. This system allows operators to maintain a safe distance between themselves and the drill string without the need for cumbersome and unreliable umbilical cables. In addition, a proximity detection system was developed for the shaft jumbo and introduced at the Ernest Henry shaft. The system detected when any person was within 1 m of the drill feed and shutdown the machine. This barrier system reduced the exposure of personnel working on the shaft bench to rotating drill strings and therefore reduces the overall risk of the drilling cycle.

Ground Support

Background

Ground support in shafts has traditionally been a labour intensive manual task and surface support has either been a form of steel mesh (chainlink or weld mesh) or hand sprayed shotcrete. Both methods are time consuming and introduce hazards to the operation. Steel mesh requires personnel to work outside the sinking stage handrails to install it against the shaft walls and is subject to blast damage, whilst hand sprayed fibrecrete exposes operators to dust, fumes and pressurised hoses.

Question

Can a spray robot similar to that used in lateral development be adapted for shaft use, and provide hydroscaling for scat control?

Solution

A shaft specific shotcrete robot was designed and manufactured for temporary ground support and bench blow over. The robot was introduced at the Ernest Henry Mine in 2013 and enables the shaft sinker to hydroscale then spray a temporary membrane of fibre-reinforced shotcrete to the shaft walls prior to personnel rock bolting the shaft walls. The robot has also been designed with the capability to blow over the shaft bench, thus removing the requirement for manual handling of a 50 mm diameter compressed air blow pipe (Figure 8).

Shaft excavation

Background

Shafts have traditionally been excavated using either a 'Cryderman' style clamshell vertical shaft mucker or a 'cactus grab' style lashing unit. These are pneumatically driven machines which rely heavily on the skill of the operator to manoeuvre and position the jaws/grab in the muck pile and not interfere with personnel working on the shaft bench during the mucking cycle.

A prototype hydraulic excavator style shaft mucker was previously designed to optimise productivity and advance rates, whilst improving operator safety for the Kencana K2 shaft, as all previous shafts at Kencana had been excavated by hand. This significantly reduced manual handling issues and increased productivity. However, whilst superior to hand mucking, the prototype was inefficient compared to traditional Vertical Shaft Muckers (Cryderman, Herman, Brutus etc) or cactus-grab lashing units and not practical for large diameter shafts. In large diameter shafts, multiple kibbles are often required, with personnel on shaft bottom changing full and empty kibbles to optimise the winder hoisting system. Both cactus-grab lashing units and pneumatic vertical shaft muckers depend on the skill of the operator to maximise

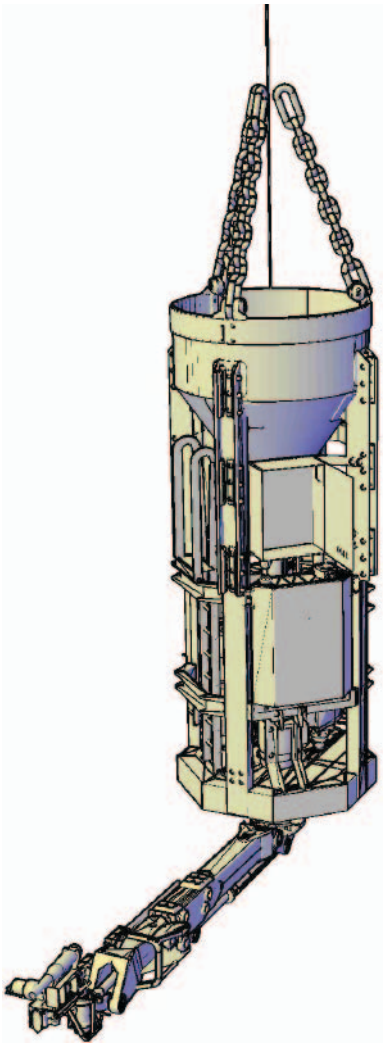


FIG 8 – Shaft shotcrete machine.

efficiency and minimise risk to personnel on the shaft bottom, which is a soft risk control for a hazardous activity.

Question

Can a hydraulic steering system be designed for pneumatic shaft muckers to improve precision and control without affecting the simplicity and ease of use of the machine?

Solution

An advanced system for shaft muckers was designed and manufactured which gives precise control for positioning of the mucking unit and results in a more efficient mucking cycle and greater safety for personnel. This system was effectively used at Ernest Henry. A fully operational 'spare' unit is set-up for training of all operators on surface prior to operation underground. Whereas vertical shaft mucker operators in North America are highly skilled and experienced personnel, it was possible to train 'green' operators to an acceptable level of proficiency within one week on the hydraulic shaft mucker, thus providing a safer and controlled work environment at the shaft bottom (Figure 9).

Fatigue monitoring and identification

Problem

Fatigue related events contribute to many incidents in the underground mine. This leads to significant damage, downtime and in extreme cases injury. Various strategies and systems have been trialled and investigated in the field of fatigue management. The majority of these systems have proven impractical due to sensors requiring constant focus on the operator and as a rule, are reactive systems, ie they provide notification that an event has occurred. The ultimate goal of fatigue management is to be able to predict when an employee is likely to be fatigued and thus stop the event from occurring!

However, we as an industry do not have a baseline for an optimum operator environment, or bluntly, we don't know what the root cause of fatigue is.

Question

Do environmental factors such as oxygen levels, heat, humidity and background vibration contribute to operator fatigue?

Solution

In cooperation with Sano Health and Edith Cowan University PhD candidates, Byrnecut have commenced a program to assess environmental conditions and relate these to operator fatigue. This study will assess known fatigue events (such as micro-sleeps) using existing logging devices in terms of environmental conditions noise, vibration, humidity, CO, CO_2, O_2, heat). Our aim is to correlate these findings with operator alertness (as measured by biometric sensors) with a view to developing a method of predicting events before they happen.

The findings of this study will be presented in an alternate paper and have potential to change how our industry identifies and manages fatigue.

Hybrid truck technology

Background

Surface mining trucks have used diesel-electric drivetrains dating back to the 1920s (Aggregates Business Europe, 2009), the technology has been proven in this environment (Brown, Elbacher and Koellner, 2000; Koellner *et al*, 2004) and offers the following benefits to underground haulage:



FIG 9 – Shaft mucker.

- higher top speed
- high starting torque
- low maintenance.

Although higher top speeds are not as applicable to underground trucks, the high starting torque will result in smoother performance on a changing grade and less speed changes compared to a mechanical drive truck. As underground declines have varying grades and surface conditions, mechanical drive trucks tend to hunt between gears when encountering changes in gradient, this causes a significant loss of momentum and thus forward speed. The electric traction motors with maximum torque from stall are not affected in the same way and are able to maintain ground speed across changing road conditions. Whilst trolley assisted electric trucks have been used in underground applications, these are limited to specific mines with the capability to install an overhead trolley line, hybrid drives have the potential to offer the benefit of electric drive without the trolley infrastructure.

Question

Will a hybrid diesel-electric truck offer greater productivity and reliability in an underground environment compared to a mechanical drive truck.

Solution

Byrnecut have commenced a research project with the Edith Cowan University School of Engineering in Perth to investigate the use of electric drive technology for mine haul trucks. This project engages automotive engineering students to develop a haulage truck concept using four electric wheel motors to maximise available payload capacity for a minimal footprint.

CONCLUSIONS

The underground mining sector in Australia is one of the most technologically advanced and competitive mining environments on the planet. Australian contractors are expanding operations overseas and enhancing mine safety, mechanisation and productivity across the globe. The domestic industry has created opportunities for contractors to grow, develop and flourish since they first appeared on the WA Goldfields in the 1980s. Contractors have developed systems, resources and technology that are world-class and available to any mine owner. Through challenging industry accepted practices and pursuing any competitive advantage, this has created a disruptive force which continually leads to improved operating practices and a safer and more efficient mining industry.

Mining contractors are a valuable resource to the mining industry – use them, challenge them and reap the rewards.

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